Comparative study of biomass chemical looping gasification over CuO and NiO supported on olivine synthesized by different temperature

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Abstract: The oxygen carrier (OC), maintaining a long lifetime with better reactivity, becomes the key to the Biomass chemical looping gasification (BCLG). In this study, CuO and NiO supported on olivine (Cu/Ni/O) were synthesized at different temperature (900, 1100 and 1300°C). Characterization methods (XRD, SEM, BET, H_2-TPR, TG and TG-FTIR) have been adapted to investigate the effect of calcination temperature on the component, morphology, multiple redox, oxygen carrying capacity and activity of the OCs. XRD shows that $CuFe_2O_4$ and $NiFe_2O_4$ phases are observed after copper and nickel loaded onto peridot. The surface area and oxygen-carrying capacity of the OC are reduced from 4.760 m^2/g and 3.60 wt.% at 900 °C to 1.305 m^2/g and 1.93 wt.% at 1300 °C, respectively. Correspondingly, the final weight loss of OCs with CS also showed the same trend as redox process of OCs, and two weightless intervals emerged. The anti-resistance of the OC has greatly improved, among which 1100-Cu/Ni/O shows the better anti-resistance with attrition rate of 0.09 wt.% and pulverization rate of 0.41 wt.%. BCLG experiments were conducted in a micro fluidized bed reactor (MFBR) to evaluate the performance of OCs. Under the optimal operation conditions (mass ratio of OC to biomass = 30:1, steam injection rate = 0.05 g/min and temperature = 800°C), the highest gasification efficiency of 59.2% and syngas yield of 0.71 Nm^3/kg were obtained over 1100-Cu/Ni/O. Cyclic test demonstrated that well sintering resistance and cyclic performance were observed by 1100-Cu/Ni/O, which made it very attractive for the biomass gasification process.

Keywords: Biomass Chemical looping gasification; micro-fluidized bed reactor; Cu/Ni/olivine; Anti-attrition; Reactivity.

1. Introduction

Among the biomass utilization technologies, chemical looping gasification (CLG) stands out as the one with the lowest energy and economic cost in converting biomass into syngas (H_2, CO) or H_2 rich gas[1-3]. The conventional gasification agents are replaced by lattice oxygen which is generated from Oxygen carrier (OC). In a typical CLG system, OC is circulated between an air reactor and a fuel reactor. In the fuel reactor, biomass reacts with OC and is converted to H_2, CO, CH_4 as well as CO_2. After contributing the lattice oxygen, OC is reduced to a low valence oxide or simple substances. The reduced OC, together with the unconverted biomass fractions (char), are fed into the air reactor, to heat up the OC and burn off the char. The reduced OC is oxidized again into high valence oxide when contacting with oxygen, subsequently, transmitted into the fuel reactor to provide the heat necessary for gasification[4-6].

To date, much of the research has focused on finding the appropriate OC for the process since it is one of the keystones in the development of the CLG technology. Low cost gives natural ores a critical advantage over other candidates. As a magnesium-iron-silicate, olivine, has been extensively tested in the laboratory and large-scale experiments owing to its satisfying crushing strength, abrasion resistance and catalytic activity[8]. Large-scale tests has been carried out to evaluate the performance of olivine in the 2–4 MWth indirect biomass gasification unit using wood chips [9], in which 30% tar destruction was obtained by olivine as bed material. Ryd´en et al[10] assessed the crushing behavior of 25 materials by a spray cup, founding that OCs were more likely to exhibit longer lifetime and reactivity when the mechanical strengths of OC was higher than 2N. Olivine, ranging from 180 to 250 μm, was measured by Wang et al [11], who found that the crushing strength of calcined olivine reached 5.6N which was higher that of iron ore(3.1N) [12]. In addition to catalytic effect and excellent mechanical strength, olivine shows a certain oxygen-carrying capacity. A redox test of olivine has been carried out in TGA to quantify oxygen transport capacity by providing alternating reducing/oxidizing atmosphere (H_2 as reducing agent and air as oxidizing agent). As a result, 0.5 wt.% of oxygen was obtained [13]. Same test has been conducted in our previous work, and the result showed that 1.07wt.% of lattice oxygen was acquired from olivine in H_2+N_2 atmosphere [14].

Reactivity of OC must be tailored so sufficient fuel conversion and syngas yield can be obtained in the fuel reactor. However, natural ore, often has low reactivity during continuous redox reactions. One of the most effective means of mitigating this disadvantage is the modification of natural ores [15]. Compared to single metal oxide, the synthesized bimetallic OCs have shown better synergistic properties owing to bimetallic OCs can effectively make up for the shortcomings of single metal oxide [16]. It is generally recognized that Fe, Ni, Cu, Co, Mn, Ca and Ti et al. are the common modifiers in the chemical looping system. Recently, the above pure metal oxides have been combined in pairs to form mixed metal oxides based on the physicochemical properties of these metal oxides. Tian et al. [17] have conducted CLG experiments using cement bonded OC (Cu20Fe80@C) which displayed superior redox reactivity in fluidized bed after 30 cyclic with lignite. Approximately 100% carbon conversion and 75% H_2 yield were achieved by the Ni and Ce modified olivine which sustained for 70 h without deactivation [18]. The behavior of bimetallic Fe–Ni OC was assessed in a 10 kWth unit using sawdust pine as fuel, and compared to the Fe_2O_3/Al_2O_3, a higher gasification efficiency of biomass was obtained[19]. But to our knowledge, little is known from previous studies on the investigation of bimetallic Cu/Ni in CLG. The Cu-based OC has an excellent oxygen uncoupling capacity, which provides an ideal oxygen source for CLG, but, in terms of tar destruction, the catalytic performance of Cu-based OC is unremarkable[20]. Among other active components, Ni-based OC exhibits promising performance for tar cracking[21]. Based on this, by combining the oxygen uncoupling property of CuO, tar destruction capacity of NiO and abrasion resistance of olivine, a bimetallic Cu/Ni/olivine has been synthesized (CuO and NiO loaded on olivine) in our previous work, exhibiting better redox and catalytic performance for biomass gasification [14].

During the CLG process, the OC circulates between fuel reactor and air reactor continuously. Therefore, the mechanical strength and lifetime of the OC is critical to determining the long-term operation of the CLG. There are many factors in reducing the lifetime of OCs, among which the invalidation of OCs caused by attrition is one of the crucial problems. Attrition of OC was inevitably caused by CLG units during the process of cycling, including a high-speed riser, a fluidized bed, a cyclone separator, a low-velocity bubbling bed and a couple of loop seals. In addition, the thermal expansion of the particles caused by the temperature gradient and chemical reactions, such as abrasion, fracture, disintegration and splitting, deteriorate the capability of the OC [22]. A test carried out in a bench scale fluidized bed shown that structure, maturation and microstructure of $CuFeAlO_4$ and $CuFe_{1.5}Al_{0.5}O_4$ can play a major role in physical attrition propensity [23]. The lifetime of CU-4 OC was tested by Cabello et al.[24], founding that the operating temperature had a great impact on the lifetime of OC. The lifetime of CU-4 OC decreased from 5000 hours at 800 °C to 450 hours at 900℃. The attrition of OC has been tested by Liu et al.[22] under different condition, including cold attrition experiment, hot attrition experiment and reactive attrition experiment, in order to quantify the contribution of stress to OC's attrition, such as mechanical stress, thermal stress and chemical stress. Based on the change of morphology, the distribution of surface elements as well as the crystalline phase of the OC, they found that the chemical stress greatly affected the performance of the OC as the number of cycles increases.

Much effort has been spent on increasing the abrasion resistant behavior of OC, where temperature-enhanced method has been widely used to strengthen the integration between the active component and the carrier. The effects of the calcination temperature on the OC have been conducted in a 30 kWth chemical looping system, and a remarkable finding is that the OC, calcinated at higher temperature, displayed better strength and attrition performance [25]. After calcined at higher temperature, a strong interaction between Cu and olivine was detected in Cu/olivine, among which the 3% Cu/olivine, prepared at 1000°C, did not deactivate in the oxidation-reduction cycle, displaying the same property during 10 cycles [26].

The same results were found by Meng et al.[27], who found that part of the Fe was fused into the inner structure of olivine forming a new (Mg,Fe)Fe_2O_4 phase, and a higher H_2 yield as well as a stronger resistance to carbon deposition were found in the case of Fe/olivine calcined at 1400℃. Based on this, it is feasible to enhance the integration between the active component with carriers by increasing the temperature of calcination, thus enhancing the anti-attrition properties of the OC.

The objective of the present paper is to improve the attrition resistance of OCs while maintaining their activity. Therefore, this research designs three types of Cu/Ni/O prepared by different temperature of calcination to compare the changes in composition, morphology and property. X-ray diffraction (XRD), Scanning electron microscopy (SEM), temperature programmed reduction (TPR) and Brunauer–Emmett–Teller (BET) were performed to confirm the phase and the micromorphology changes of the OC after calcination at different temperature. The redox property of the Cu/Ni/O was investigated by redox thermogravimetric analysis (TGA). The CLG performance of the OCs were conducted in a micro fluidized bed reactor (MFBR) in terms of the carbon conversion, total gas yield, lower heating value as well as gasification efficiency.

2. Experimental section

2.1 Biomass and OC preparation

The CS is obtained from Xinjiang province, China, and used as biomass in the present study. The CS, a mean size between 100 and 125μm, was collected through the operation of crushing and screening. The moisture of sample was removed after dried at 105°C in drying ovens. Proximate and ultimate analysis of CS listed in Table 1.

Table1 Proximate and ultimate analysis of cotton stalk

Proximate analysis (wt.%, ad)				Ultimate analysis (wt.%, daf)					LHV(MJ/kg)
Moisture	Ash	Volatiles	Fixed carbon[a]	C	H	O[a]	N	S	
4.35	6.23	73.88	15.55	48.20	6.40	43.97	1.06	0.37	17.06[b]

a:by difference b: calculated based on Dulong equation

Wet impregnation method was adopted to prepare the Cu/Ni/O. At the first, the raw olivine particle size with 100~150 μm calcined at 900°C for 4h. And then, the calcined olivine, $Cu(NO_3)_2\cdot 9H_2O$ and $Ni(NO_3)_2\cdot 6H_2O$ with a mass ratio of 9:6:85 mixed in deionized water for 12h. It is worth noting that an ultrasonic environment was provided during the evaporation of water by the rotary evaporator, allowing for a more homogeneous preparation of the particles. A small amount of water remaining in the sample was further dried in an oven at 110°C after rotary evaporation. At end of preparation, the samples were calcined in a muffle furnace at 900, 1100 and 1300°C for 4 h, and named as 900-Cu/Ni/O, 1100-Cu/Ni/O, 1300-Cu/Ni/O respectively.

2.2. Apparatus and Operational Conditions.

Performance of CLG of CS with Cu/Ni/O as OC evaluated in a MFBR (Fig.1). MFBR is equipped with an electromagnetic pulse system, a bubbling fluidized bed, a syringe pump, and an on-line mass spectrometer (AMETEK, USA) for feeding the CS, gasifying the CS, supplying the steam, and detecting the gases, respectively. The quartz tube reactor incorporates K-type thermocouple and is divided into three zones by two porous plates, where the gasification reaction takes place in the middle zone (with a height of 40 mm and an inner diameter of 20 mm). Prior to the experiment, different mass of OC (1, 2, 3 and 4g) was loaded into the middle zone, and then heated up to the desired temperature (700, 750, 800 and 850°C) in Ar atmosphere (purity of 99.999%) at a flow rate of 500 ml/min. Then a certain flow of steam (0~0.075 g/min) is passed through, 1 minute prior to sampling, to provide a H-rich environment. Once steady-state conditions were reached, 100 ± 0.3 mg of CS was immediately sprayed into the reaction zone. After cooling the unreacted tar and water at -15°C, the dry tar-free gas (H_2, CH_4, CO, CO_2) released from the gasification is measured by means of an on-line mass spectrometer. The produced gas is collected in 5-minute periods using gas sample bag, and is detected by Micro-GC (INFUCON). The end of the reaction is judged by whether the signal lines of the individual gases on the mass spectrum return to the baseline. After that, O_2+Ar (1:4) is introduced into the reactor to oxidize the OC.

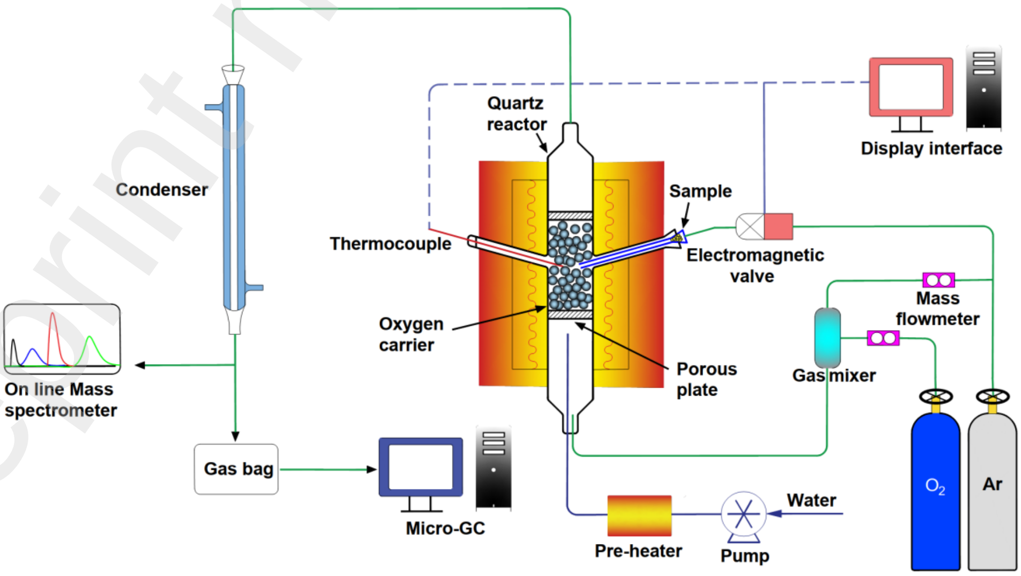


Fig. 1. Diagram of the MFBRA

2.3. Thermogravimetric analysis (TGA)

2.3.1 Thermogravimetric multiple redox experiments

A 10-cyclic isothermal reduction-oxidation experiments of the prepared OCs were conducted in a TGA (SDT Q600, TA Instruments) at 800°C, to evaluate the oxygen carrying behavior and stability of OCs. Prior to each TGA test, about 30 ± 0.2 mg of OC, supported on a ceramic crucible, was heated in air atmosphere up to 800°C at a heating rate of 10°C/min. The 10%H_2 + 90%N_2, providing reduction atmosphere, was passed in the reduction stage with a reduction time of precisely 15 minutes. After the reduction step, the OC was exposed to air for 10 min in the oxidation stage. It is worth noting that the contact of H_2 and O_2 at the border of the oxidation stage and reduction stage has been avoided by N_2 which was introduced as purging gas for 5 min. Mass flow meter controller is used to ensure that the gas is passed through at a flow rate of 100 mL/min at each stage.

2.3.2 Thermogravimetric analysis coupled to infrared spectroscopy

The non-isothermal reaction between OCs and CS were quantitatively determined in a TG-FTIR (TGA 8000, Perkin Elmer), which was capable of real-time weighing the changes of mass and measuring the gasification products. Mixture of CS (4 ±0.2 mg) and OCs (40 ±0.2 mg) were heated from 40 °C to 900 °C (ramp rate of 30 °C/min), and then kept isothermal for about 10 min. Prior to each test, the adapter inlet, transfer line and TG-FTIR cell are heated from ambient temperature to 270 °C, avoiding the condensation of products. FTIR spectra were obtained continuously in the range of 4000-500 cm^{-1} with resolution of 2 cm^{-1}.

2.4. Characterization of OC

X-ray diffraction (XRD) patterns of the obtained samples were measured with SmartLab (Rigaku Co, Japan) using a Cu Ka source operating typically at 40 kV. Microstructures of neat and modified OCs and their compositions were observed by SEM and EDX (Hitachi Regulus8100). Hydrogen temperature-programmed reduction (H_2-TPR) was performed on a MICROMERITICS AutoChem II 292. The specific surface areas of the samples were determined by the Brunauer–Emmett–Teller (BET) method with nitrogen using a AUTOSORB IQ instrument (Quantachrome Instruments, USA).

2.5 Data evaluation

The performance of the OC was evaluated based on the variation of four main gas components. On the basis of the total volume of gas, the volume of each gas can be obtained. Eq.1 and Eq.2 were used to calculate the total volume of gas and volume of each gas, respectively:

$$V_{out} = \frac{V_{Ar}}{1 - \sum X_i)} \quad (1)$$

$$V_i = V_{out} \times X_i \quad (2)$$

where V_{out} is the total gas volume (m^3), obtained by applying a certain flow rate of Ar. X_i were expressed as the volume fractions of i (i = H_2, CH_4, CO and CO_2) in the gas products (including Ar). Vi is the volume of gas i (m^3).

Carbon conversion to gas is the ratio of the carbon present in CS converted into carbon-containing gas products, which shown as follows:

$$\text{Carbon conversion to gas} = \frac{(V_{CO} + V_{CO_2} + V_{CH_4})/V_m \times 12}{\text{Mass of carbon in the sample}} \times 100\% \quad (3)$$

Total gas yield (Nm^3/kg) was estimated based on the ratio of the total volume gas to the mass of CS, which was calculated by Eq. (4):

$$\text{Total gas yield} = \frac{V_{H_2} + V_{CO} + V_{CO_2} + V_{CH_4}}{\text{Mass of the sample}} \quad (4)$$

The lower heating value (Mj/kg) of gas was based on the content of the combustible gases and calculated by Eq. (5):

$$Q_{LHV} = \frac{108_{\varphi H_2} + 126_{\varphi CO} + 359_{\varphi CH_4}}{1000} \quad (5)$$

where symbol φH_2, φCO, and φCH_4 (%) means the content of the H_2, CO and CH_4 respectively, in the gas products without Ar.

Gasification efficiency, η (%), was calculated by Eq. (6):

$$\eta = \frac{Q_{LHV} \times \text{Total gas yield}}{Q_{CS}} \quad (6)$$

where Q_{CS} (MJ/kg) is express as the total lower heating value of CS (listed in table 1).

The oxygen transport capacity of OC (Ro), obtained by weighing the mass, was estimated as:

$$R_o = \frac{m_{ox} - m_{red}}{m_{ox}} \times 100\% \quad (7)$$

where m_{ox} denotes the mass of fully oxidized particles and m_{red} in the reduce form.

2.6. Attrition test

In this study, a standard vibrating screen has been used to carried out the attrition tests, according to the GB/T10504–2017, China [18]. Prior to the test, crushed powder in the OCs is removed by a standard screen (pore size = 0.25 mm). Approximately 100 g of OC (size ranging from 0.45 to 0.85 mm), noted as m_1, is placed on a vibrating sieve and shaken continuously for 3 hours, weighing at 30 minutes intervals. When the vibration was completed, two standard screens, with pore size of 0.25 mm and 0.15 mm, are used to separate broken particles and abrasive powders, which are noted as m_2 and m_3, respectively. The attrition rate and pulverization ratio has been evaluated by the mass change of sample:

$$\text{Attrition rate} = \frac{m_2}{m_1} \tag{8}$$

$$\text{Pulverization ratio} = \frac{m_3}{m_1} \tag{9}$$

3.Results and discussion

3.1 Characterization of fresh OC

Fig.2 (a) shows the XRD analysis of the carriers prepared at different temperatures. As seen, the main diffraction peaks of calcined olivine were Mg_2SiO_4, SiO_2 as well as Fe_2O_3, and R-1 and R-2 may occur during calcination. However, the diffraction peak of Fe_2O_3 disappears, while $MgFe_2O_4$ and a strong diffraction peak of Mg_2SiO_3 is found at 1300°C, possibly occurring the R-3. Fig.2 (b) shows the effect of temperature on the olivine after loading CuO and NiO. Diffraction peaks of NiO and CuO were both detected at all three temperatures, where the peak of NiO tended to decrease significantly with increasing temperature, while the diffraction peak of CuO did not change significantly. In addition, it can be noted that $CuFe_2O_4$ has better peaks out at 1100°C. A strong diffraction peak of $NiFe_2O_4$ is detected in 1300-Cu/Ni/O. Meanwhile, Fe_2O_3 is disappeared. The data indicated that NiO and CuO not only load in the form of oxides on the surface of olivine, but also react with Fe_2O_3 formed new phases of $NiFe_2O_4$ and $CuFe_2O_4$ (as listed in R-4). Further observation reveals that the Mg_2SiO_3, which originally appeared in the olivine calcined at 1300 °C, has a significantly smaller peak intensity in Fig.2(b), and tends to decrease with increasing preparation temperature. Moreover, the intensity of the diffraction peak of SiO_2 also decreases and disappears at 1300 °C. It is possible that Mg_2SiO_3 and SiO_2 react further to form Mg_2SiO_4 at higher temperature (as listed in R-5).

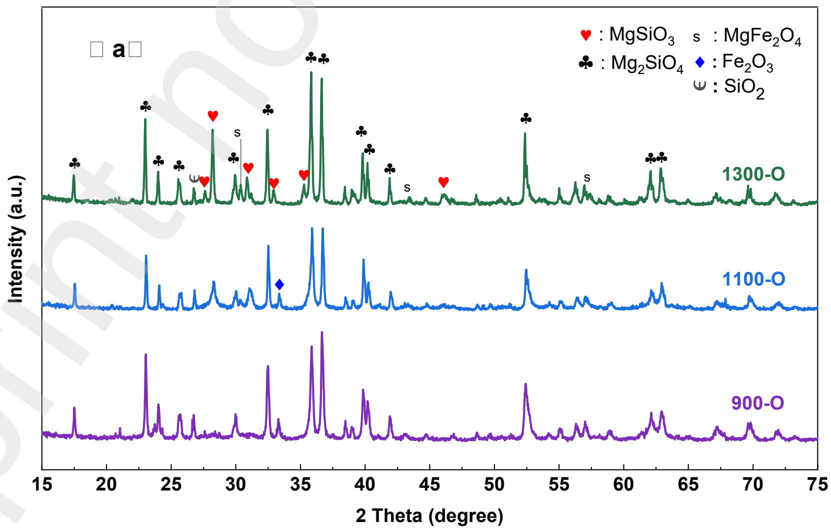
$$(Mg,Fe)_2SiO_4 + O_2 = Mg_2SiO_4 + SiO_2 + Fe_2O_3 \quad \text{(R-1)}$$

$$Mg_2SiO_4 + SiO_2 = 2MgSiO_3 \quad \text{(R-2)}$$

$$Fe_2O_3 + MgSiO_3 = MgFe_2O_4 + SiO_2 \quad \text{(R-3)}$$

$$NiO/CuO + Fe_2O_3 = Ni/CuFe_2O_4 \quad \text{(R-4)}$$

$$SiO_2 + Mg_2SiO_3 + O_2 = Mg_2SiO_4 \quad \text{(R-5)}$$



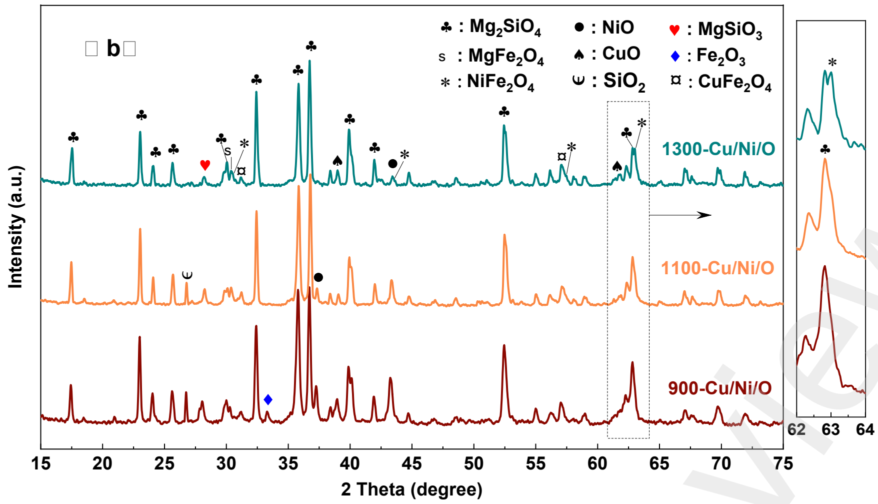


Fig. 2. XRD spectra of fresh OCs. (a): olivine calcinated at different temperature; (b): Cu/Ni/O calcinated at different temperature.

The surface morphology of the OCs is shown in Fig. 3. According to the SEM images of the OCs, the surface of the 900-Cu/Ni/O is relatively compact with a uniform and full particle, which appears to be more microporous. As the calcination temperature increases, it can be seen that the porous structure of the 1100-Cu/Ni/O gradually disappears and that the surface particles gradually form large particle groups. When the calcination temperature is increased to 1300°C, the pore structure of the1300-Cu/Ni/O largely disappears and becomes relatively smooth, and particles appear sporadically on the surface. A homogeneous distribution of Fe, Ni and Cu on the outer surface of OCs is observed, as well as a proper integration of these elements inside the support.

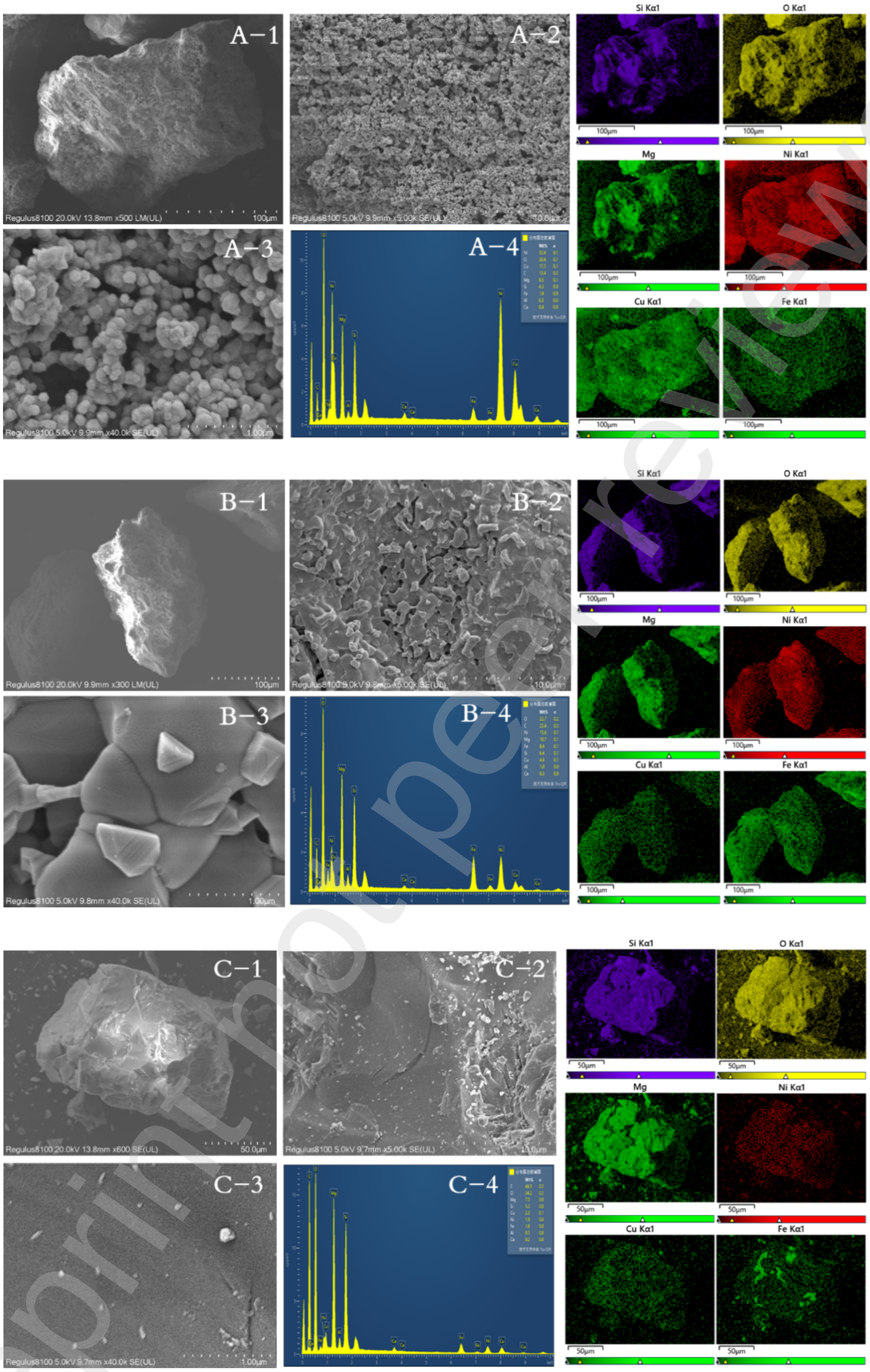


Fig. 3. SEM images of the fresh OCs. (A): 900-Cu/Ni/O; (B): 1100-Cu/Ni/O; (C): 1300-Cu/Ni/O; (1-3): Magnification; (4): EDS of OCs.

The surface area and average pore diameter of OCs are listed in Table 2. With a specific surface area only of 1.666 m^2/g, olivine, calcinated at 900℃, can be defined as little or none-pore structure. The specific surface area and pore volume of the 900-Cu/Ni/O are 4.760 m^2/g and 3.524×10^{-2} cm^3/g, respectively, which are 3 and 10 times higher than that of the 1100-olivine, respectively. As the calcination temperature rises, the specific surface area and pore volume of 1100-Cu/Ni/O and 1300-Cu/Ni/O decrease, of which the pore structure is also almost nonexistent in 1300-Cu/Ni/O. Correspondingly, the average pore diameter is almost negligible in 1300-Cu/Ni/O. This phenomenon can be attributed to the fact that, at the lower calcination temperature, the CuO and NiO are adhered to the olivine resulting in an increase in the specific surface area, some of which may fuse into the pore structure to block the pores and cause the average pore diameter to decrease. Nevertheless, the metal loaded on the surface migrates further to the inner surface at higher temperatures on account of the low melting point of some oxides, such as copper (1083℃).

Table 2 BET analysis of the investigated oxygen carriers

Samples	Surface area (m^2/g)	Pore volume (cm^3/g)	Average pore Diameter(nm)
900-Olivine	1.666	8.763×10^{-3}	74.653
900-Cu/Ni/O	4.760	3.524×10^{-2}	33.895
1100-Cu/Ni/O	1.575	3.313×10^{-3}	10.373
1300-Cu/Ni/O	1.305	2.398×10^{-3}	-

The H_2-TPR was employed to have a greater knowledge of the reducibility of the prepared OCs. As can be seen from Fig. 4, three reduction peaks near 530°C, 582°C and 753 °C are found for olivine calcined at 900, 1100 and 1300 °C, respectively, where the reduction peak is more obvious for 900-olivine, and there is no clear boundary between the two reduction peaks for 1100-olivine. The temperature of the reduction peak of olivine calcination at 1100 and 1300 °C is shifted towards higher temperatures. Combined with XRD and SEM, this phenomenon can be explained by the fact that the Fe_2O_3 on the olivine surface was drastically reduced, and some Fe elements were present as $MgFe_2O_4$ within the olivine structure, making the high-valent Fe are difficult to reduce [14]. In the case of 900-Cu/Ni/O, three peaks have been detected at 253℃，507℃ and 826℃. The first one is representing the reduction of CuO. The second can be attributed to the reduction of NiO and free iron oxides. The third one can be ascribed to the reduction of $MgFe_2O_4$ as well as the other spinel or mixture of Cu and Ni oxides inside the olivine. Compared with the 900-Cu/Ni/O, there is a large shoulder peak near 540°C in 1100-Cu/Ni/O, which can be attributed to the reduction of $Fe_2O_4^{2-}$ as well as the mixture of iron and nickel oxides or spinel that is present inside the grain. For the Cu/Ni/O prepared at 1300℃, the peak near 574℃ becomes wider and smaller than that of 900-Cu/Ni/O and 1100-Cu/Ni/O.

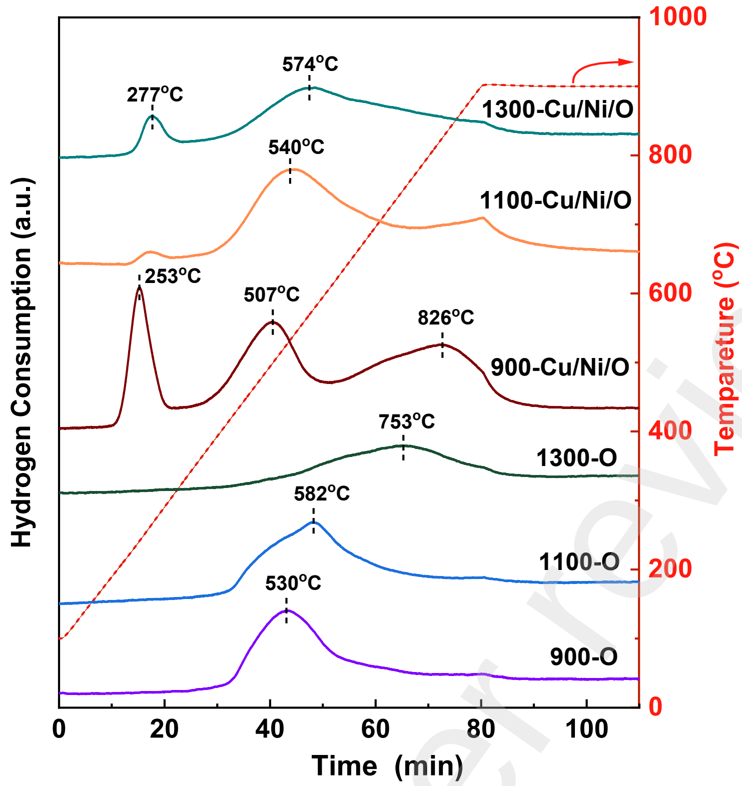


Fig. 4. H_2-TPR profiles of OCs

3.2 The redox behaviors of OCs evaluated by TGA

Utilizing oxygen released from thermal decomposition of OCs has been proposed for gasification of solid fuels, thus, enhancing the efficiency of gasification. To determine the oxygen transport ability of OC, 10-cycle decomposition–reoxidation reaction of 900-Cu/Ni/O, 1100-Cu/Ni/O and 1300-Cu/Ni/O were conducted in TGA at 800°C with hydrogen for decomposition and air for oxidation. The plots of relative weight loss versus time of various OCs and a total 10 redox cycles are shown in Fig. 5. It can be found that the weight loss of OCs maintained a stabilized state after first three redox cycles, suggesting that all of the Cu/Ni/O seemingly need an activation process to achieve an optimum redox state. As the circulation proceeds, the oxidation degree of the OCs is gradually degraded. Zhao et al.[28] tested the redox behavior of $NiO/NiAl_2O_4$ on TGA, founding that a small weight loss of the OC occurred with increasing redox times. They speculated that this was mainly related to the deactivation of nickel oxide. The loss of mass would not be due to the decomposition of the OCs, which have been calcined at a higher temperature that were higher than the redox temperature of TGA. This could be caused by the inability to oxidize some of the sub-surface material such as Cu_2O and Fe_3O_4. Moreover, there shows a decreasing trend of the oxygen donating capacity with the increase of calcination temperature.

Experimental data of the OCs at the 8th cycle has been chosen to further investigate the redox behaviors of OCs. The Ro value of 900-Cu/Ni/O, 1100-Cu/Ni/O and 1300-Cu/Ni/O were 3.60 wt.%, 2.40 wt.% and 1.93 wt.%, respectively, at a same mass percentage. Two main reduction periods were found in the reduction profile of the OCs, namely fast and slow reduction period. For the 900-Cu/Ni/O, mass reduction of 2.90 wt.% has been occurred in the fast reduction period, which accounts for 80.6% of the Ro of 900-Cu/Ni/O. This period may attribute to the property of oxygen uncoupling of copper oxide ($CuO = Cu_2O + O_2$) during the nitrogen purging stage. However, only 0.70 wt.% of weight loss has been observed during the hydrogen purging stage, which may be due to partial decomposition of Fe_2O_3 to Fe_3O_4 or FeO. It should be noted that the weight of OCs gained rapidly for OCs, indicating that the oxidation process of OC occurs faster compared to the reduction process. After stabilization, the reduction rate of 900-Cu/Ni/O was about 1.25 %/min, while the oxidation reaction rates are about 6.25 %/min, which is 5 times higher than that of the reduction rate.

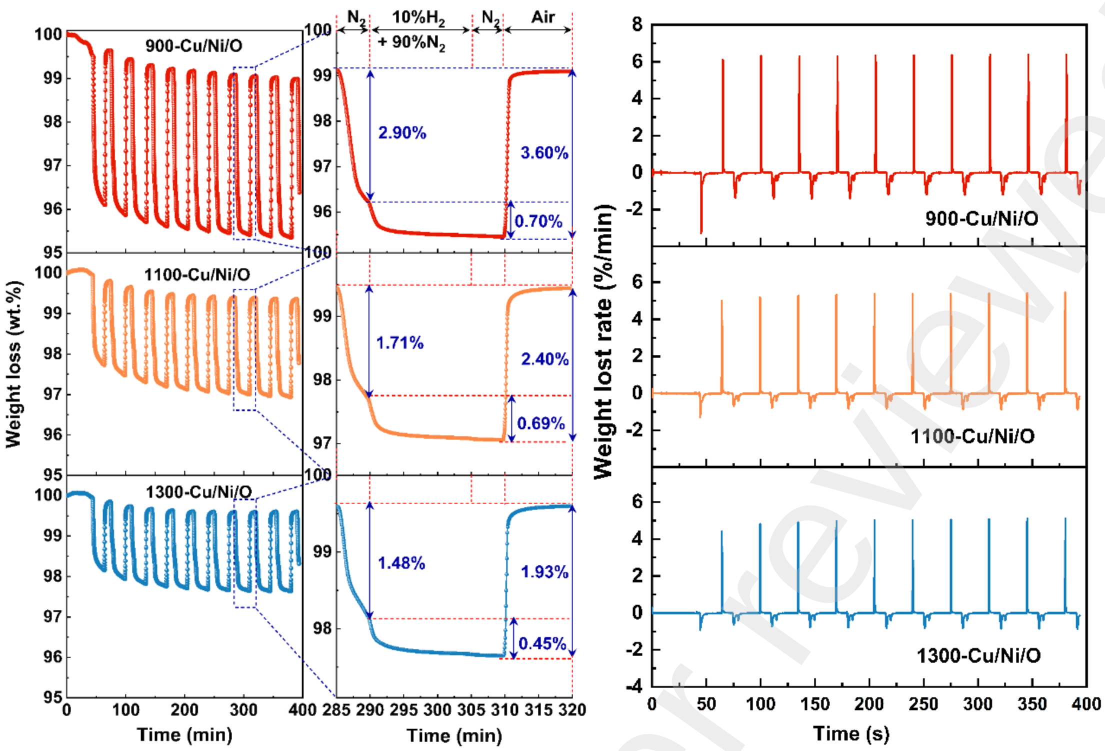


Fig. 5. Multiple TGA redox cycles of OCs at 800 °C.

3.3. Attrition tests

One of the advantages of olivine compared to dolomite and other nature ore is that olivine has strong abrasion resistance. Fig. 6(a) and (b) illustrated the variation of attrition rate and pulverize rate over time. It can be observed that the pulverulent rate of carrier and OCs are increased with the increasing time. Moreover, pulverulent rate of 900-olivine, 1100-olivine, 1300-olivine are all significantly higher than that of raw-olivine (pulverulent rate was 0.14 wt.%), which can be attributed to the fact that the temperature has changed the main crystal phases of olivine, resulted in a diminishment in mechanical strength. This phenomenon agrees with Meng's observations, in which calcined olivine also displayed the same trend [18]. It can be found that the attrition rate also becomes greater with time, particularly with the 900-Olivine. However, there is also a decrease in the attrition rate with time, mainly due to further collisions of the large worn particles breaking up into smaller ones. However, the OCs loaded with active components showed a different trend in terms of pulverization rate, where the pulverization rate of 900-Cu/Ni/O reached a maximum of 1.13 wt.% after 3h, leading to the loss of active metals. Excitingly, the 1100-Cu/Ni/O showed the lowest pulverization rate (0.41 wt.%), which is even smaller than the pulverization rate of olivine after 3 hours, indicating that higher temperature enhances the interaction between the active metals and the olivine, and promotes the reorganization of the active metals into the olivine structure. In addition, 1100-Cu/Ni/O shows a better anti-resistance ability, making it possible to use in a fluidized bed. The pulverization of the OC is mainly caused by friction between OC particles and OC particle surfaces as well as low velocity collisions between OC particles and the reactor wall, mainly occurring in the dense zone of the reactor. Bulk fracture of the OC occurs mainly in the dilute region of the reactor, where the particles undergo fragmentation due to the higher collision energy, resulting in some coarse fragments[29]. As a result, the cracks at the impact site further propagate to the particles and produce some non-elutriation particles.

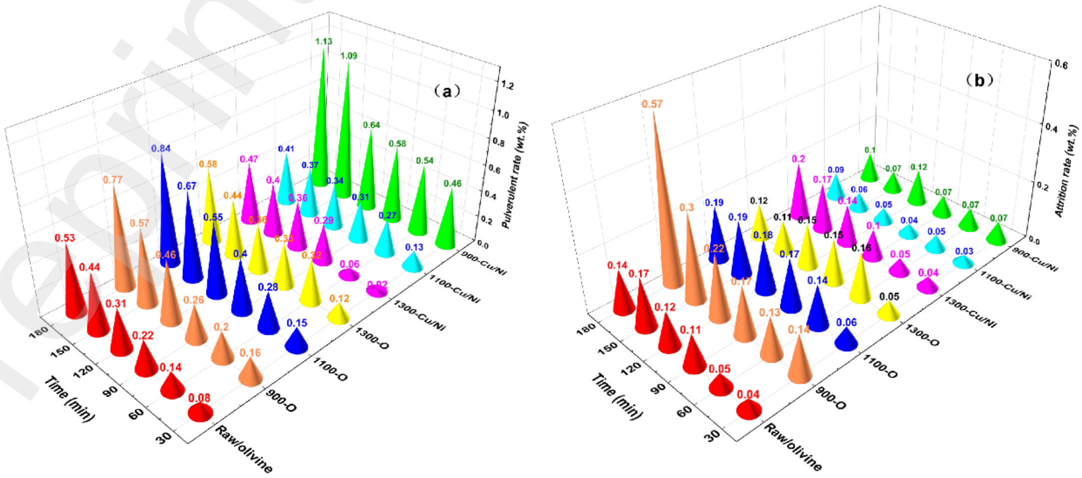


Fig. 6. Attrition test of OCs. (a): Pulverization rate; (b): Attrition rate.

3.4 Non-isothermal CLG characteristics in TG-FTIR

To better ensure the effect of OCs on CS, the non-isothermal CLG properties of CS and OCs were evaluated using TG-FTIR (shown as Fig. 7). It could be observed from the DTG profile that two individual reaction zones, occurring in sequence, were detected. The first reaction zone (zone 1), occurring at the temperature from 300 to 600 °C, was attributed to the emission of volatiles with fast weight loss, where maximum weight loss 5.90 wt.% was obtained in 900-Cu/Ni/O and 1100-Cu/Ni/O at the same time. However, the maximum weight loss rate was discovered in 1300-Cu/Ni/O with a weight loss rate of 1.32 %/min. The second zone (zone 2), extended from 600 to 900 °C, was associated with gasification of char left after the zone 1. Particularly, weight loss rate up to 0.3 %/min was detected in zone 2, which was much lower than that at the zone 1, especially for 1300-Cu/Ni/O with CS. Three peaks located at 742.5 °C, 750.6 °C and 761.1 °C were observed in the DTG profile of 900-Cu/Ni/O, 1100-Cu/Ni/O and 1300-Cu/Ni/O, respectively, which can be related to the reaction between OCs and residual char. Such a result confirmed that the reaction of 900-Cu/Ni/O with char was earlier than of other two OCs. The TG curves showed that, after full gasification, the mass loss of 900-Cu/Ni/O reached around 8.71wt.%, which was higher than that of other two OCs, indicating that thermochemical reactivity of 900-Cu/Ni/O was higher than that of 1100-Cu/Ni/O and 1300-Cu/Ni/O during gasification.

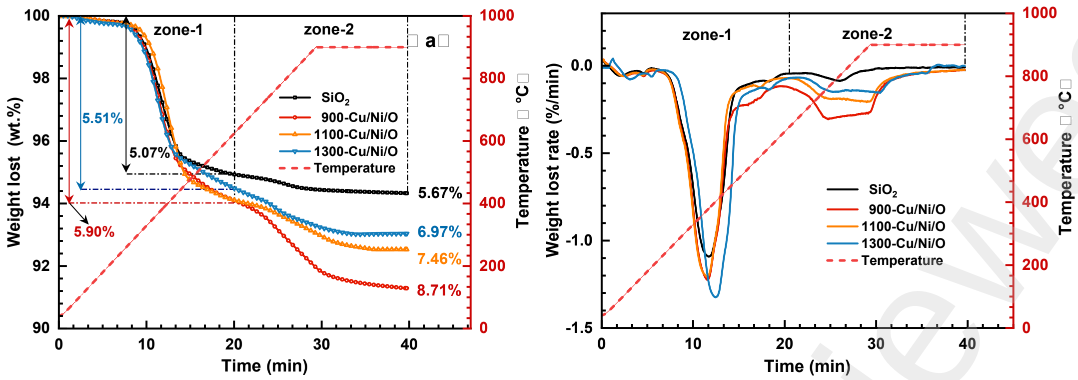


Fig. 7. Weight lost and Weight lost rate curves for CLG of CS over Cu/Ni/O synthesized by different temperature.

Fig.8 presents 3D-FTIR images of the products generated from thermal decomposition of CS. On the basis of the characteristic absorbance values, the pyrolysis products mainly contain H_2O (4000-3500cm^{-1}), CH_4 (3000-2900cm^{-1}), CO_2 (2400–2240 cm^{-1} and 667 cm^{-1}), CO (2250-2060cm^{-1}), C-H(1743cm^{-1}), C-C and C-O (1085cm^{-1}). It can be seen that the distributions of gaseous products are different with different OCs. Peaks in the range 2400-2224 cm^{-1} are represented to the bending of C=O, stretching asymmetrically to form CO_2. In the case of OCs, slight increase on CO_2 intensity has been shown compared to SiO_2, while the absorbance intensity of C=O, C-C, C-O and C-H compounds decreased. The wave of CO_2 at 2357 has been singled out and shown in Fig. S1, making it easier to analyze the impact of OCs on biomass. It is remarkable that two peaks appear in the formation process of CO_2, which is corresponding to the thermogravimetric process. CO_2 began to evolve around 300 °C, and reached the maximum intensity at 432 °C during the zone 1, in which the impact of 900-Cu/Ni/O is more evident. To go a step further, the increase in CO_2 intensity could be due to the presence of CuO, NiO and Fe_2O_3 according to the data of

H_2-TPR, which releases lattice oxygen at low temperature. The same trend for CO_2 is found out in zone 2, where mainly occur the reactions between char and some oxides hard to reduce in OCs.

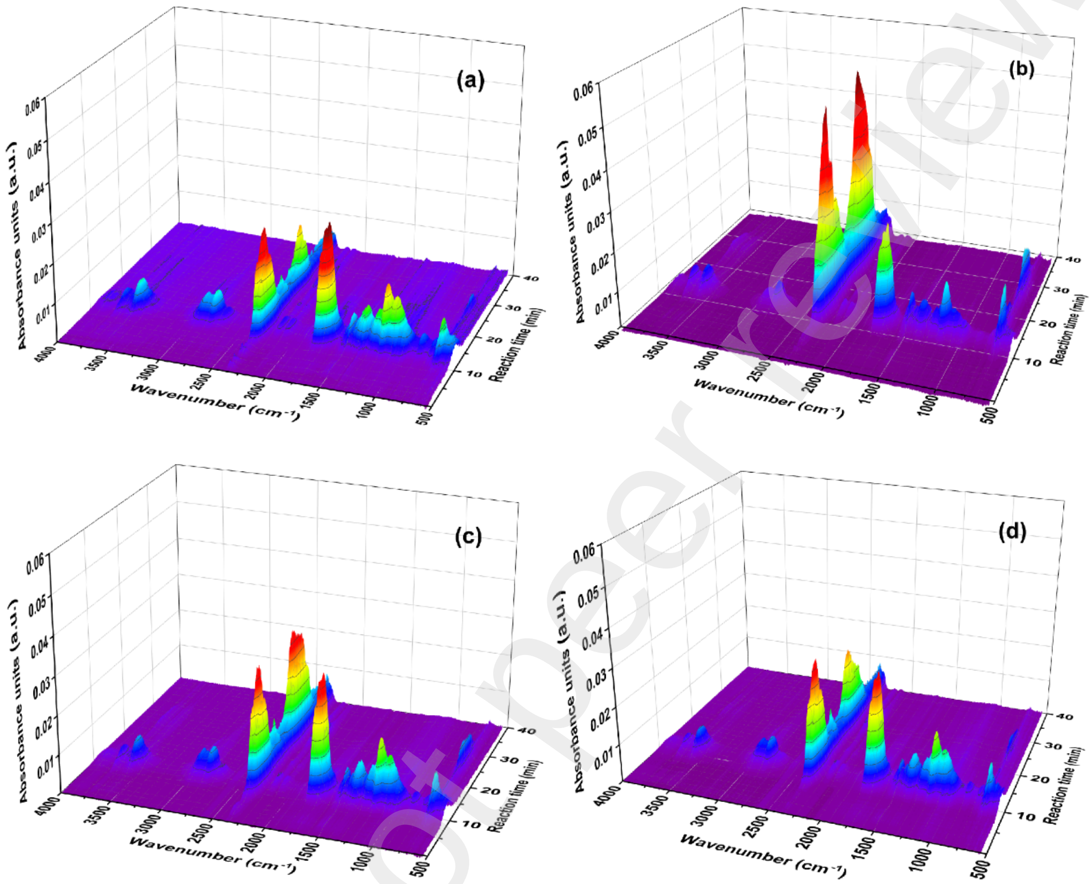


Fig. 8. 3D-FTIR diagram of gaseous products under (a) SiO_2; (b) 900-Cu/Ni/O ;(c) 1100-Cu/Ni/O; (d) 1300-Cu/Ni/O

3.5 Effect of experimental parameters of CLG in MFBR

Samples with different OC to biomass ratios (OC/B) were tested to evaluate the amount of OC on the gasification of CS. As illustrated in Fig. 9, compared to the inert material (SiO_2), upgrading of the biomass-derived gas over the 900-Cu/Ni/O led to an increase in the yield of CO, H_2 and CO_2, and a decrease in the yield of CH_4, where CO (0.19 to 0.24 Nm^3/kg) and H_2 (0.09 to 0.13 Nm^3/kg) reached the maximum at the ratio of 30:1, whereas CO_2 (0.09 to 0.29 Nm^3/kg) reached the maximum at the ratio of 40:1. The total gas yield and carbon conversion increased from 0.43 Nm^3/kg to 0.67 Nm^3/kg and 40.2% to 64.4%, respectively, as the OC/B increased from 0 to 30. Nevertheless, a small decrease on the total gas yield was obtained as the ratio of OC/B increased to 40. A conclusion can be draw that appropriate OC/B, providing lattice oxygen and active sites for gasification of biomass, was instrumental in the conversion of sample. As a result, an increment of the gas yield, carbon conversion and gasification efficiency were acquired [30]. However, higher ratio of OC/B impacts considerably the producer gas composition, as a result of blocking of the contact of steam and pyrolysis products, including bio-char and volatiles [31]. Moreover, a part of CO and H_2 can be oxidized into CO_2 and H_2O since the excessive oxygen donated by OC [32].

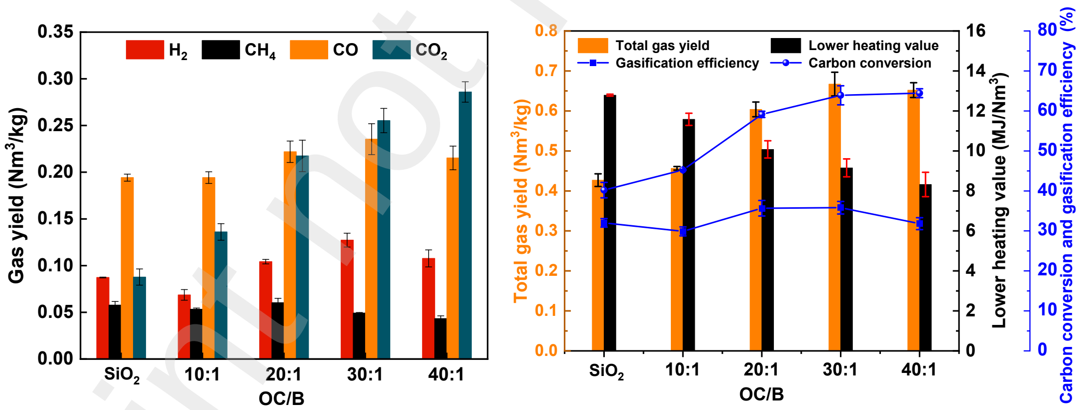


Fig. 9. The effect of oxygen carrier to biomass ratio (OC/B) on CLG of CS over 900-Cu/Ni/O. Gasification temperature:800 °C

It is well accepted that carbon conversion and syngas production can evidently be promoted by means of elevating gasification temperature. The influence of temperature, ranging from 700 to 850 °C, on comparison of gas component over

1100-Cu/Ni/O is presented in Fig. 10. The yield of H_2, CO and CO_2 increased as the temperature rising, while an opposite tendency was observed for the CH_4. Meanwhile, the total gas yield, carbon conversion and gasification efficiency grew up from 0.42 Nm^3/kg, 41.9% and 30.3% to 0.73 Nm^3/kg, 64.5% and 44.6%, respectively. Nevertheless, the trend in carbon conversion and gasification efficiency at 850 °C is minimal compared to 800 °C. With strong heat and mass transfer, a series of complex and heterogeneous chemical reactions occurred when CS pulse entered the reaction zone. Chemical bonds are more likely to break in high temperature, allowing the thermal cleavage of large molecule compounds into smaller molecule gases. In addition, the Ro of the OC is enhanced at high temperatures, providing more lattice oxygen. Besides, the high temperature accelerates the flow rate inside the reactor, resulting in an increase of the frequency of effective collisions between the OC and the CS.

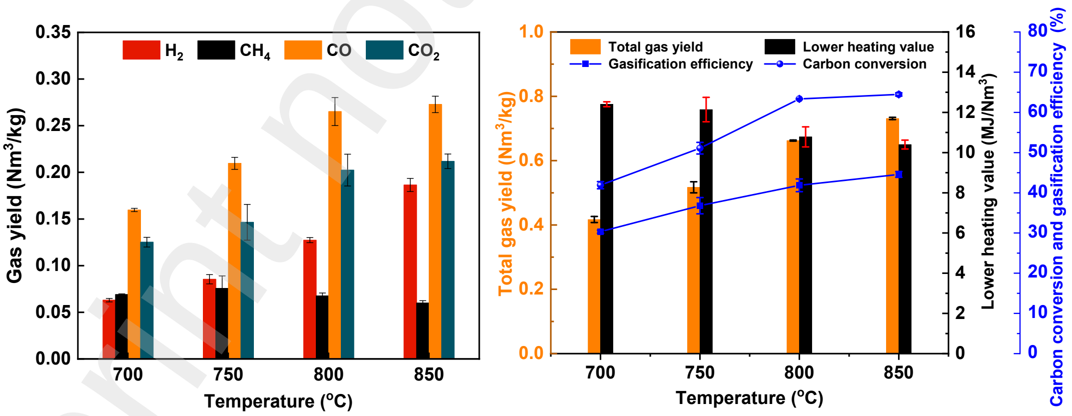


Fig. 10. The effect of gasification temperature on CLG of CS. OC: 1100-Cu/Ni/O; OC/B ratio =30:1.

The steam, providing hydrogen source except the biomass, exert a positive momentous effect on BCLG. As show in Fig. 11, steam injection rate, varying from

0.025 to 0.075 g/min, was analyzed to ensure the appropriate ratio of steam so as to promote the production of hydrogen-rich gas and carbon conversion. It can be seen that a significantly increasement appears in the yield of H_2, up from 0.13 Nm^3/kg to a maximum yield of 0.35 Nm^3/kg with a steam injection rate of 0.075 ml/min. It means that the H_2 yield increased by 269% compared with no steam injection, proving that the addition of stream is beneficial to hydrogen production. However, the yield of CH_4, CO and CO_2 were firstly increased with the increasing of stream injection rate from 0.025 to 0.050 g/min, then, displayed a downward trend. It can be observed that the gas yield, carbon conversion and gasification efficiency are significantly improved after adding steam. The results show that the carbon conversion rate, gas yield, gasification efficiency and lower heating value have been significantly improved with the addition of steam, reaching the maximum value at 0.050 g/min with the value of 81.0%, 1.02Nm³/kg, 59.2% and 9.91MJ/kg, respectively. The fact demonstrated that the char gasification reaction, and methane reforming reaction have been promoted in the presence of steam.

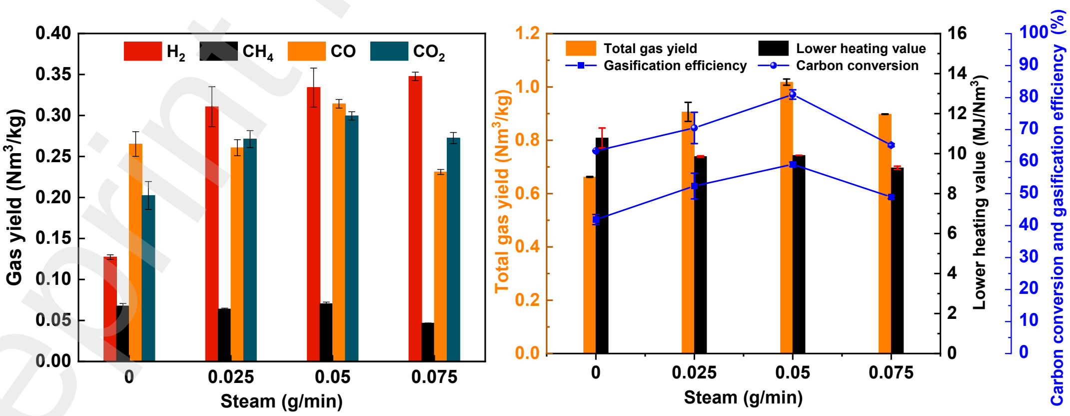


Fig. 11. The effect of steam flowrate on the gas production from CLG of CS. OC: 1100-Cu/Ni/O; OC/B ratio =30:1; gasification temperature:800 °C.

The generation process of the four main gases was tracked in real time by mass spectrometry to further compare the reaction process of the four main gases during SiO_2, 1100-Cu/Ni/O and 1100-Cu/Ni/O-steam. Generally, there are two main reactions occurred in BCLG. Firstly, oxides in OC are reacted with Pyro-gas (including tar, CO, H_2, CO_2, CH_4 and so on) which are thermally decomposed from biomass, reducing into low valence metal oxides or metal monomers after the devotion of lattice oxygen. Then, solid-solid reaction ([O] + C= CO/CO_2) occurs in high temperature, usually, taking longer time to finish and thus determining the reaction. However, as is depicted in Fig. 12, it can be seen that the reaction between 1100-Cu/Ni/O and CS is finished within 50s which is same as that of SiO_2, verifying the OC could release lattice oxygen in a very short time. Moreover, in the case of 1100-Cu/Ni/O, the intensity of these gases are higher than that of pyrolysis, which is consistent with Fig. 9. Additionally, CO_2 is the main product in the combustion stage ($Ar+O_2$), mainly generated from the combustion of the remaining char. The CO_2 produced by combustion after gasification is slightly less than that produced by combustion after pyrolysis. According to TGA redox experiments, 3 g of 1100-Cu/Ni/O can theoretically provide 0.072 g of lattice oxygen, which is less than the oxygen equivalent required for the totally gasification of CS (1.46 g/g_{CS}). It can be concluded that, in the absence of steam, the reaction of OC with CS is completed in a very short time, and that the OC reacts mainly with the Pyro-gas as well as has less effect on the char. There was a decrease in intensity of these gases after adding steam to the system, whereas the reaction time increased significantly, dividing gasification process into two stages, a fast reaction stage and a slow reaction stage. In the fast reaction stage, a large amount of Pyro-gas fraction reacts with the OC and steam to produce small gaseous molecules. Multiple reactions, including char reacted with the steam and OCs to produce H_2, CO and CO_2, and re-oxidation of OC by steam to produce H_2, can take place during the slow reaction zone.

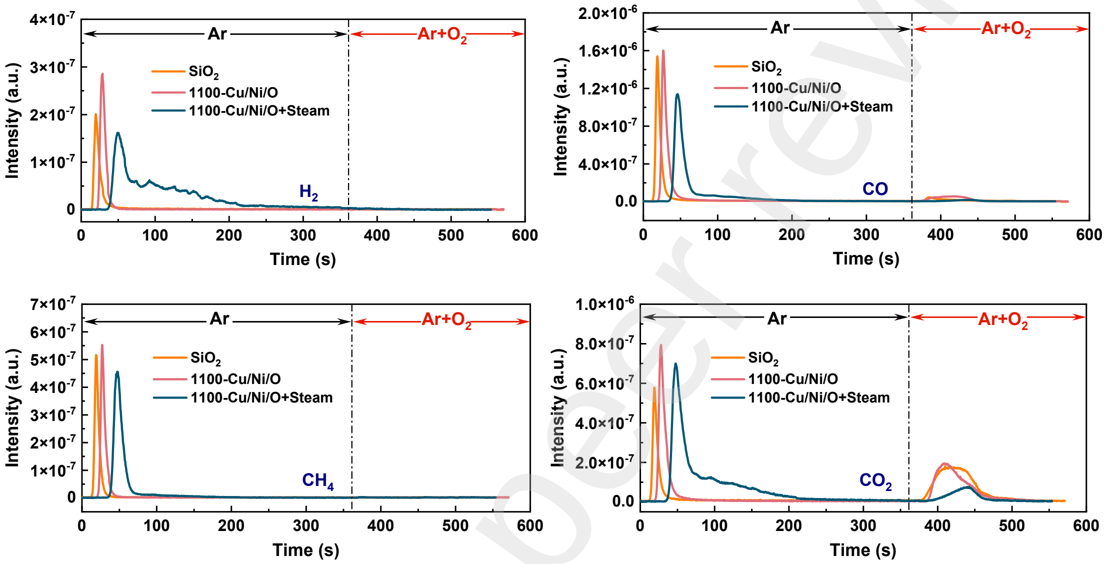


Fig. 12. Real-time changes of H_2, CH_4, CO and CO_2 during CLG of CS.

The anti-resistance of the OC is greatly improved by calcination temperature, but it is unknown whether the activity of OCs during gasification will be enhanced by preparation temperatures. Therefore, a comparative study of OCs on the CLG performance of the CS were performed according to the above optimized conditions. As illustrated in Fig. 13, the yield of H_2 increased from 0.30 Nm^3/kg to 0.36 Nm^3/kg, while the gas yields of CO_2 decreased from 0.45 Nm^3/kg to 0.27 Nm^3/kg with the calcination temperature increased. Besides, as the calcination temperature rising, the total gas yield deduced from 1.04 Nm^3/kg to 0.91 Nm^3/kg and the carbon conversion rate decreased from 87.4% to 64.5%. The decreasing CO_2 can be attributed to the different Ro of OCs, combined with the characterization analysis of the TGA which decreases with increasing temperature of calcination. Correspondingly, carbon conversion has also been significantly reduced. It should be noted that the 1100-Cu/Ni/O has the highest CH_4 and CO production with 0.07 Nm^3/kg and 0.31 Nm^3/kg, respectively. Besides, in addition to producing better syngas yield, better gasification efficiency and lower heating value can also be obtained when 1100-Cu/Ni/O is present during steam gasification of CS. The highest hydrogen yield was obtained in the presence of 1300-Cu/Ni/O, but total gas yield and carbon conversion of 1300-Cu/Ni/O were much smaller than those of the other two OCs. The $NiFe_2O_4$, formed in the 1300-Cu/Ni/O, has been reported according to the literature that can further promote the cracking of intermediate product in the biomass gasification[33, 34]. Nevertheless, smaller specific surface area and Ro of 1300-Cu/Ni/O limit its further performance.

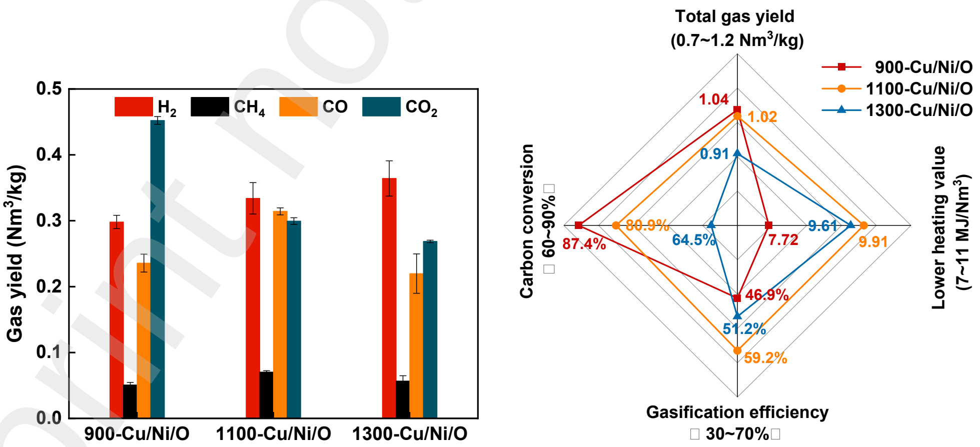


Fig. 13. The effect of OCs on the gas production from CLG of CS. (OC/B ratio =30:1; Gasification temperature:800 °C; Steam flowrate: 0.05 g/min.)

10 redox cycles for CLG of CS have been carried out in the MFBRA in the cause of evaluating the uncoupling ability and stability of the 1100-Cu/Ni/O. Overall, as

illustrated in Fig. 14, H_2 and CO_2 yield fluctuated and then stabilized after 4 cycles, while a slightly downward trend in CO yield was observed as the number of cycles increases. Similarly, the total gas yield and carbon conversion maintained near 1.05Nm^3/kg and 80% respectively, during subsequent cycles, indicating that 1100-Cu/Ni/O exhibited better reactivity during multiple CLG reaction. This phenomenon can be attributed to the structure and reactivity of the OC tended to stabilize with the cycle proceeding, which was in agree with the result shown in Fig. 5. It further proves that the OCs requires an activation process to achieve a stabilized redox capability during CLG process.

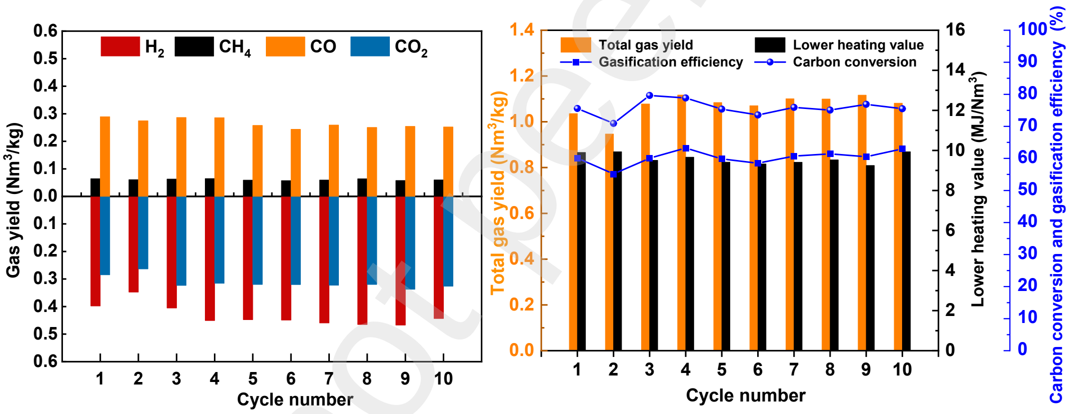


Fig. 14. Cyclic performance tests of CLG of CS over 1100-Cu/Ni/O. (OC/B ratio =30:1; Gasification temperature:800 °C; Steam flowrate: 0.05 g/min.)

XRD analysis of OCs after 3-cycle fluidized bed test and the fines collected during the 10-cycle test are shown in Fig. S2. However, metal monomers or low valence oxides of iron, nickel and copper have not been detected in Fig. S2, indicating that the reduced OCs, in the presence of steam, would be further oxidized. Besides, XRD of the sample after the 10th reaction was very similar to that of the fresh sample.

Fig. S3 shows the SEM micrographs of the OCs after 3 cycles and after 10 cycles. It can be noticed that small bubble holes appear on the particles of 900-Cu/Ni/O after the reaction, while the smooth surface of 1300-Cu/Ni/O becomes rugged after three reactions and fine grooves appear. After three reactions fine particles appeared on the surface of 1100-Cu/Ni/O, and, subsequently, small cluster of particles were formed after 10 reactions. The elemental mapping of the pellets after reaction showed that Cu and Ni are well-distributed in the pellet. Besides, potassium was detected on the surface of 1100-Cu/Ni/O after the tenth reaction in EDS analysis.

4. Conclusions

The main goal behind the present investigation was to raise the performances of olivine used into BCLG by adding copper oxides and nickel oxides. The study was focalized on optimization and characterization of OC in term of the anti-resistance and reactivity by means of an increase in calcination temperature. The effect of the preparation temperature on the composition, morphology, redox properties, oxygen transport capacity, attrition rate and activity of the OCs is clarified.

X-ray diffraction shows that the Mg_2SiO_4 phase was the main component of the calcinated olivine, and $CuFe_2O_4$ as well as $NiFe_2O_4$ phases were observed after copper and nickel loaded onto peridot. Based on SEM/EDX and BET analysis, it was concluded that CuO as well as NiO gradually melted from the outer surface layer into the inner surface layer of the olivine as the calcination temperature increases. Attrition resistance of OCs is determined using the fines collected during the standard vibrating screen tests, which are in the order of 1100-Cu/Ni/O > 1300-Cu/Ni/O > 900-Cu/Ni/O. In the multiple TGA redox tests, oxygen transport capacity of 3.60%, 2.40 % and

1.93% were obtained for 900-Cu/Ni/O, 1100-Cu/Ni/O and 1300-Cu/Ni/O, respectively. Correspondingly, the final weight loss of OCs with CS also showed the same trend as redox process of OCs. The infrared absorption peaks of CO_2 in the product are particularly distinct, with the larger the oxygen-carrying capacity the more CO_2 is produced by the OC.

In MFBR tests, the amount of OC used in the gasifier strongly affects the performance of the CLG process, leading to upward trend in total gas yield as well as carbon conversion efficiency and a downward trend in lower heating valve, but excess amount of OC will further consume H_2 and CO to form H_2O and CO_2. The increment of temperature significantly enhances the gas yield and the process efficiency. The main variable affecting to the process performance parameters was the steam injection ratio, where the content of H_2 has been trending upward with the steam amount used, reaching a peak of about 0.35 Nm^3/kg for the 1100-Cu/Ni/O at steam injection ratio of 0.075 g/min. Additionally, higher steam injection ratio also enhances the carbon conversion, the gas yield and gasification efficiency. However, excess steam may cause a decline in the BCLG performance due to heat loss in the gasifier. Interestingly to note that the calcination temperature strongly influences the reactivity of OCs in the BCLG. Under optimal operating conditions, 1100-Cu/Ni/O gives averagely a better result in terms of CO production with a greater gas yield of 0.31 Nm^3/kg as well as higher gasification efficiencies of 59.2%, while the carbon conversion of the 1100-Cu/Ni/O is lower than that of 900-Cu/Ni/O with about 7%. A stable performance was also observed during a 10-cyle test with 1100-Cu/Ni/O, suggesting that the 1100-Cu/Ni/O has been proved as a suitable option for the chemical looping gasification of biomass.

Acknowledgements

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Fig. 14. Cyclic performance tests of CLG of CS over 1100-Cu/Ni/O. (OC/B ratio =30:1; Gasification temperature:800 ℃; Steam flowrate: 0.05 g/min.)

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